

Indian Institute of Technology Jodhpur

Subject: Time Series Analysis Problem Sheet-I

Question: Answer the following questions.

1. Let $\{Y_t\}$ be sequence of independent normal random variables, each with mean 0 and variance σ^2 . For each process check whether it is white noise or not.
 - (a) $X_t = Y_t + Y_{t-1} + 1$.
 - (b) $X_t = (\sin(t))Y_t + (\cos(t))Y_{t-2}$.
 - (c) $X_t = aY_t + bY_{t-1}$, where a and b are constants.

2. Consider the process

$$X_t = X_{t-1} + W_t$$

for $t = 0, 1, 2, \dots$, with $X_0 = 0$, where W_t is white noise with variance σ^2 .

- (a) Find the mean and autocovariance function of X_t .
 - (b) Find autocorrelation function of X_t .
3. Let $\{X_t\}$ be a time series. Compute the variance of sample mean $(X_1 + X_2 + X_3 + \dots + X_n)/n$ in terms of $\gamma_X(h)$.
 4. Consider a time series $\{x_t\}$ given by

t	1	2	3	4	5	6	7	8	9	10
x_t	1	1	-1	1	-1	-1	-1	1	-1	1

Using the above series, construct following time series

$$y_t = 2 - x_{t-1} + 0.4x_{t-1}$$

with $x_0 = -1$. Corresponding to time series $\{y_t\}$, compute $\hat{\gamma}(2)$, $\hat{\gamma}(4)$ and $\hat{\rho}(3)$.

5. Let $\{X_t\}$ be MA(2) process given by

$$X_t = Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2},$$

where $\{Z_t\} \sim WN(0, 0.16)$. Find the autocovariance and autocorrelation function for this process when $\theta_1 = 0.3$ and $\theta_2 = 0.2$ at lag h .

6. Let $\{X_t\}$ be AR(1) process given by

$$X_t = \theta_1 X_{t-1} + Z_t,$$

where $\{Z_t\} \sim WN(0, 0.25)$. Find the autocorrelation function for this process when $\theta_1 = 0.9$ at lag h .

7. For each of following, determine whether the series is stationary or not.
 - (a) $X_t = \alpha_1 + \alpha_2 t + W_t$, where α_1 and α_2 are known constants and $W_t \sim WN(0, \sigma^2)$.
 - (b) $Y_t = X_t - X_{t-1}$, where X_t is same as defined in (a).
 - (c) $X_t = Y \cos(t) + Z \sin(t)$, where Y and Z are uncorrelated random variables with mean 0 and variance 1.
8. For each of following moving average series, determine whether it is invertible or not.
 - (a) $X_t = Z_t - 0.4Z_{t-1}$.
 - (b) $X_t = Z_t - 0.3Z_{t-1} + 0.4Z_{t-2}$.
 - (c) $X_t = Z_t + 2Z_{t-1} + 4Z_{t-2}$.
9. For each of following Autoregressive series, determine whether it is stationary or not.
 - (a) $X_t = 0.8X_t + Z_t$.
 - (b) $X_t = 0.1X_{t-1} + 0.3X_{t-2} + Z_t$.
 - (c) $X_t = 2X_{t-1} + 0.5X_{t-2} + Z_t$.
10. Consider the datasets **Sunspots** and **strikes** from **itsmr** library. Plot the ACF and PACF for both datasets.
11. Consider the dataset **austourists** from **fpp2** library. Plot the dataset. Estimate its trend using smoothing with moving average filter with $q = 3$ and exponential smoothing for $\alpha=0.4$ and plot it.
12. Consider the dataset **austourists** from **fpp2** library. Compute the residuals after estimating the trend with quadratic polynomial and seasonality of 12. Using 5% level of significance, conclude whether the residuals represent independent and identically distributed noise, and determine whether they follow a Gaussian distribution